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HybridDriveVLA: Vision-Language-Action Model with Visual CoT reasoning and ToT Evaluation for Autonomous Driving

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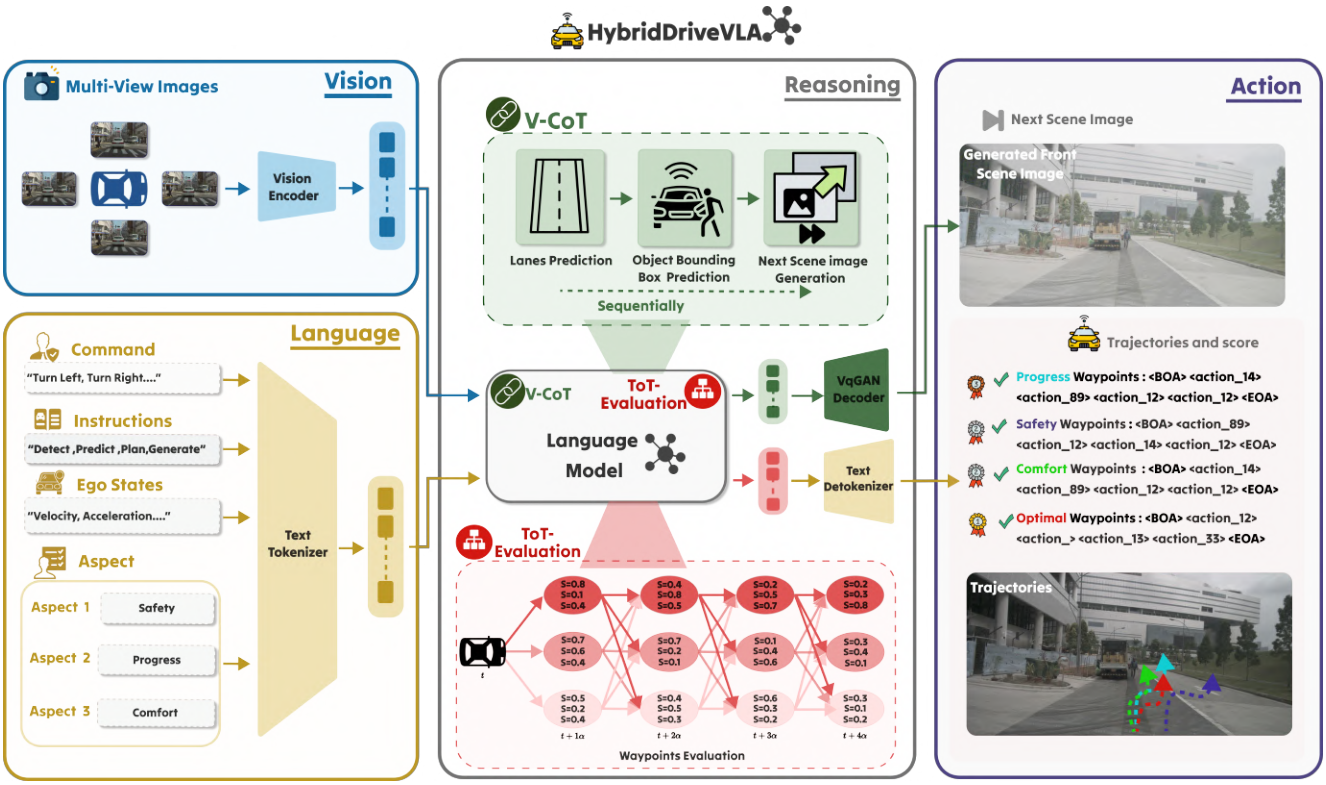


Figure 1. HybridDriveVLA is a vision–language–action model for autonomous driving that integrates visual anticipation and deliberative evaluation for selecting actions. During inference, it takes multi-view images, ego vehicle states, and instructions as input, then processes them using a vision encoder and text tokenizer. A Chain-of-Thought reasoning mechanism anticipates driving scene elements, while Tree-of-Thought evaluates candidate waypoints based on safety, progress, and comfort to select the optimal trajectory.

Abstract

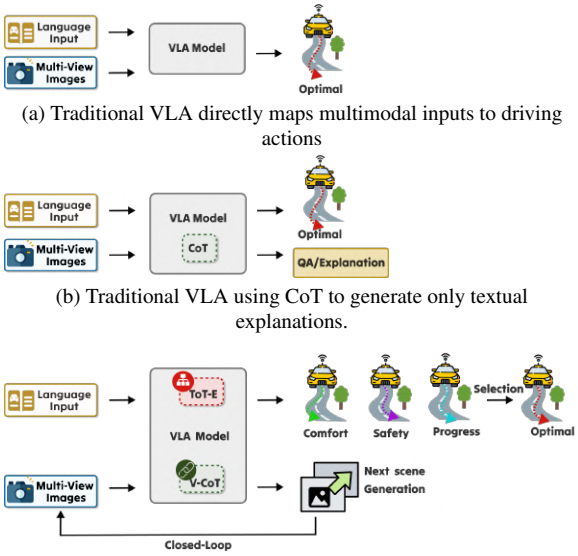
Vision-Language-Action (VLA) models are emerging as an important technology in autonomous driving, recognized for their sophisticated reasoning and interpretability. However, traditional VLA models often rely on image-to-text with Chain-of-Thought (CoT) reasoning, which converts sequential visual scenes into textual symbols, thereby underutilizing spatial context in visual information. Existing autonomous driving systems using VLA models predict only a single sequence of waypoints as a trajectory considering a given command and multiple aspects. However, we suggest

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evaluating each sequence of waypoints to reveal the importance of the corresponding aspect. We introduce HybridDriveVLA, a VLA model that integrates visual Chain-of-Thought (V-Cot) reasoning and a proposed Tree-of-Thought (ToT)-inspired waypoint evaluation (ToT-evaluation). V-Cot reasoning anticipates future scenes, which serve as goals for ToT-evaluation. The ToT-evaluation generates and scores waypoints based each on safety, progress, and comfort aspects. The highest cumulated score of the waypoints based on the three aspects is optimal. To the best of our knowledge, we are the first to propose a unified method integrating both a CoT and a ToT approach in a VLA model. Experimental results demonstrated that HybridDriveVLA achieved strong performance on comfort, progress, and safety metrics with and average collision rate of 0.17% on the nuScenes benchmark, outperforming traditional VLA models.

1. Introduction

The evolution of autonomous driving is rapidly converging towards End-to-End systems powered by Vision-Language-Action (VLA) models. VLA models [32, 34] have emerged as a promising world-model paradigm that maps raw sensor data and navigational commands directly to vehicle control actions. By leveraging their knowledge and reasoning capabilities, VLA models can interpret complex traffic scenarios and anticipate dynamic changes. Despite their advantages, traditional VLA approaches face several critical limitations that reduce their ability to reason effectively and take coherent decisions in complex driving environments. In fact, they under-utilise visual information by performing image-to-text with Chain-of-Thought (CoT) reasoning [28, 32] to improve interpretability. However, in the context of autonomous driving, traditional CoT [2, 13] that converts continuous and high-dimensional visual data into discrete textual representation, as shown in Figure 2 under-utilizes visual information. This linguistic abstraction inevitably degrades spatial information contained in visual information, which can be important for precision. Existing autonomous driving systems using VLA models predict only a single sequence of waypoints as a trajectory considering a given command and multiple aspects. However, we suggest evaluating each sequence of waypoints to reveal the importance of the corresponding aspect. Moreover, traditional VLA models [17] mostly predict only a single sequence of waypoints considering simultaneously multiple aspects, which leads to a single multi-aspect sequence of waypoints. These models predict one statistically likely sequence of waypoints that lacks individual aspect deliberation. Consequently, it may predict unsafe, uncomfortable, unprogressive waypoints especially in uncertain or multi-agent environments that require precise and flexible decision-making.



(a) Traditional VLA directly maps multimodal inputs to driving actions

(b) Traditional VLA using CoT to generate only textual explanations.

(c) HybridDriveVLA integrates V-CoT and ToT-Evaluation for visual reasoning and multi-aspect criteria evaluation.

Figure 2. Evolution from traditional VLA to our proposed HybridDriveVLA. (a) Direct action prediction. (b) CoT-based reasoning for textual explanations. (c) HybridDriveVLA as an autonomous driving world-model combining V-CoT for visual next scene anticipation/generation and ToT-Evaluation for evaluative trajectory planning on each specific aspect.

To overcome these limitations, we propose HybridDriveVLA, a framework designed to reason through visual anticipation and deliberative planning evaluation. Our method is built upon two main processes: a Visual Chain-of-Thought (V-CoT) and a proposed Tree-of-Thought-inspired waypoint evaluation (ToT-Evaluation), as shown in Figure 1. HybridDriveVLA employs V-CoT for visual reasoning by sequentially generating next scenes. This visual anticipation generates the next scenes and use those as goal state representation that preserves spatial details and continuity. Moreover, rather than relying on a single deterministic sequence of waypoints, HybridDriveVLA reasons deliberately using ToT-Evaluation to plan, generate, and score multiple waypoint sequences based on explicit safety, progress, and comfort aspects, where the highest cumulated score across these three aspects is selected as optimal. This unified integration of CoT and ToT reasonings mimics human driver's perception, anticipation, and evaluation of multiple future positions based on their safety, progress, and comfort to find the optimal trajectory [31].

To enable these reasoning abilities, HybridDriveVLA is trained on the nuScenes dataset [1]. We employ Supervised Fine-Tuning (SFT) for visual understanding and next scene generation to enable accurate V-CoT generation of future scenes. Then, Instruction Tuning (IT) to improve reason-

ing efficiency, temporal coherence, and deliberative evaluation process within ToT-Evaluation by teaching HybridDriveVLA to learn multi-path planning and aspect-based evaluation effectively. We make the following contributions:

- We propose HybridDriveVLA, an End-to-End Vision-Language-Action framework that integrates both Visual Chain-of-Thought (V-CoT) and Tree-of-Thought-inspired evaluation (ToT-Evaluation) reasoning for autonomous driving world-models.
- We design ToT-Evaluation, a deliberative multi-waypoint evaluation module that scores multiple waypoint sequences based each on explicit safety, progress, or comfort aspects and selects the optimal sequence of waypoints with the highest cumulated score.
- Experimental results demonstrated that HybridDriveVLA achieved strong performance on comfort, progress, and safety metrics on the nuScenes benchmark, outperforming traditional VLA models by achieving an average collision rate of 0.17%, demonstrating a significant improvement over existing models.

2. Related Work

Our research is situated at the intersection of End-to-End autonomous driving, large multimodal models as world models, future scene generation, and deliberative autonomous planning. This section reviews the key literature in these domains to contextualize the contributions of HybridDriveVLA.

2.1. Vision-Language-Action Models in Driving

Recent advances in Large Multimodal Models (LMMs) have led to the emergence of Vision-Language-Action (VLA) frameworks for driving [19]. These models inherit strong reasoning and world-understanding capabilities from foundation models, enabling End-to-End perception and control. Early systems such as DriveGPT4 [26] demonstrated how multimodal models could generate textual driving rationales, while DriveVLM [23], DriveLM [20], and LingoQA [14] extended this idea by incorporating textual chain-of-thought (CoT) reasoning to justify high-level decisions. However, these text-based approaches introduce a modality gap, as visual and spatial information are discretized into tokens, leading to a loss of visual information. HybridDriveVLA diverges from these methods by performing reasoning entirely within the visual domain, thereby maintaining visual continuity and enabling interpretable deliberation.

2.2. Chain-of-Thought and Tree-of-Thought Reasoning

Chain-of-Thought (CoT) prompting [32] has proven effective for structured reasoning in language models, and recent extensions into the visual domain [29] show promise for spatial reasoning tasks. In driving, however, CoT has been largely restricted to textual reasoning [16], limiting its ability to represent continuous scene evolution. Tree-of-Thought (ToT) reasoning [27] further generalizes CoT by enabling multi-path exploration and self-evaluation, a paradigm inherently suited for rich driving contexts. HybridDriveVLA introduces a hybrid reasoning process that couples a Visual CoT for anticipatory next scene generation with an Evaluative ToT for multi-aspect waypoint deliberation, thus bridging visual prediction and action evaluation in a unified loop.

2.3. Visual Prediction and End-to-End Planning

Future scene synthesis has become an emerging research frontier for autonomous driving. Generative models such as GAIA-1 [5] and DriveDreamer [25] can produce realistic future frames, yet they remain disconnected from downstream control. In parallel, End-to-End driving architectures like CIL [4], TransFuser [3], and UniAD [8] unify perception and planning but often lack explicit reasoning interpretability. HybridDriveVLA integrates its Visual Chain-of-Thought (V-CoT) to perform next-frame anticipation as part of the reasoning process itself, while the Evaluative Tree-of-Thought (ToT-Evaluation) directly grounds this visual anticipation into aspect-aware waypoint selection. This synergy enables HybridDriveVLA to anticipate, deliberate, and act in a human-like reasoning loop.

3. Methodology

The architecture of HybridDriveVLA is designed as a reasoning framework that integrates visual next scene generation and deliberative waypoint evaluation. As shown in Figure 3, HybridDriveVLA first encodes multi-modal driving inputs, performs Visual Chain-of-Thought (V-CoT) reasoning to predict the next scene visually, and subsequently applies a Tree-of-Thought-inspired waypoint evaluation (ToT-Evaluation) mechanism to deliberate across multiple sequences of waypoints before selecting an optimal sequence.

3.1. Overview

Inputs. At a current time t, HybridDriveVLA receives a set of synchronized multi-view images $I_t = \{i_t^1, i_t^2, \ldots, i_t^h\}$, the ego-vehicle state l_t, a navigational command c_t, an instruction o_t and evaluation aspect $E = \{e^{safety}, e^{progress}, e^{comfort}\}$.

Backbone VLA model. HybridDriveVLA builds upon a pre-trained backbone Vision-Language-Action (VLA) model M constituted of a vision encoder with a projector, a text tokenizer, and a text detokenizer. At the current time step t, HybridDriveVLA reasons using all inputs to

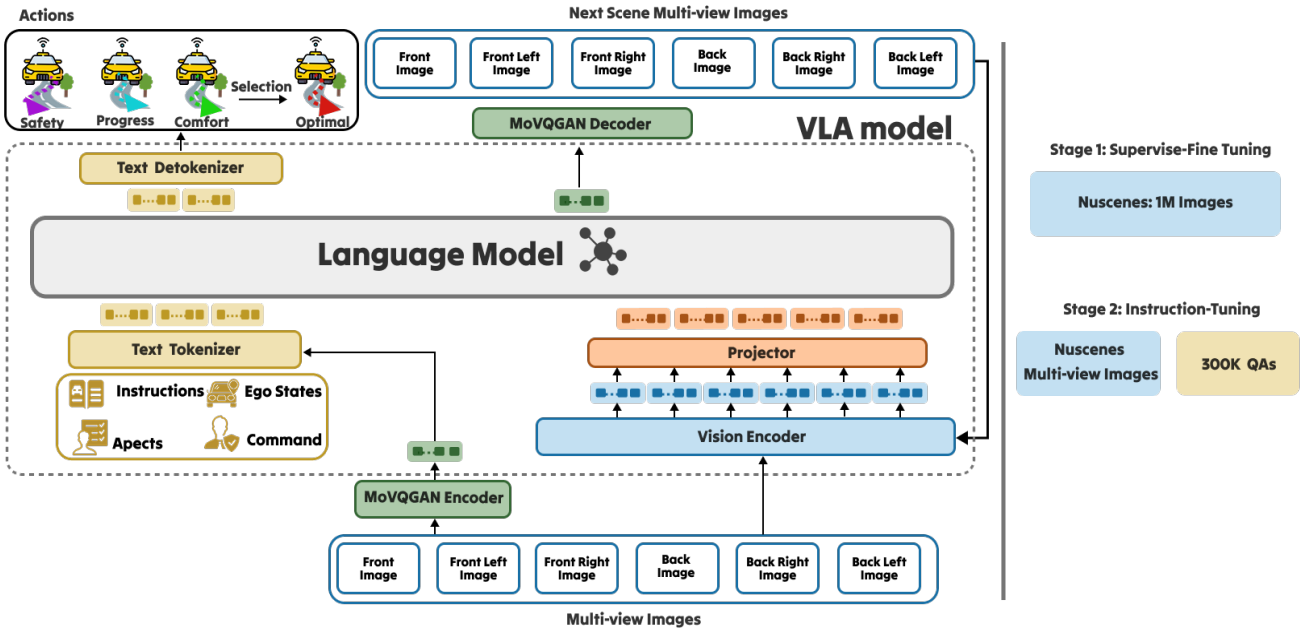


Figure 3. HybridDriveVLA architecture integrates a Vision Encoder, Language Model, MoVQGAN encoder-decoder for multimodal reasoning. Inputs include multi-view images, ego states, navigational commands, instructions, and evaluation aspects (Safety, Progress, Comfort) processed in Text Detokenizer. HybridDriveVLA outputs next scene images using MoVQGAN Decoder and sequences of waypoints as actions using a Text Detokenizer.

predict and generate autoregressively next scene output image tokens. These tokens are decoded into images using a VQGAN Decoder D_{gan} [30] to obtain the next scene image output, used as a goal to predict n multi-waypoint sequences outputs $\mathcal{A}^n$, and each waypoint set of scores $\mathcal{S}^n_{t+k\alpha} = \{s^{n,safety}_{t+k\alpha}, s^{n,progress}_{t+k\alpha}, s^{n,comfort}_{t+k\alpha}\}$.

Outputs. Conditioned on textual input features, the backbone VLA model M in HybridDriveVLA generates autoregressively the next scene image $\hat{I}_{t+6\alpha}$, and a multi-sequence waypoints $\mathcal{A}^n$ with each waypoint set of scores $\mathcal{S}^n_{t+k\alpha}$ where n is a waypoint index at time step $t + k\alpha$, k is prediction step, and α is a prediction interval between consecutive waypoints.

3.2. Visual Chain-of-Thought (V-CoT)

The Visual Chain-of-Thought (V-CoT) reasoning enables HybridDriveVLA to anticipate future scene images before planning. Unlike conventional text-based reasoning, V-CoT unfolds entirely in the visual domain, generating structured visual predictions that reflect how a human driver mentally simulates the next maneuver. Starting from the input features, HybridDriveVLA autoregressively generate next scene representation, and uses that generated representation to plan multiple waypoint sequences and their scores as a unique autoregressive sequence z_i. To incorporate explicit visual reasoning before actions generation, HybridDriveVLA first anticipates the next scene image at $t + 6\alpha$ as a goal image:

$$\hat{I}_{t+6\alpha} = M(I_t, E, l_t, c_t, o_t), \tag{1}$$

where M denotes the backbone VLA. To enable predictive visual reasoning, the Visual Chain-of-Thought (V-CoT) is trained to reconstruct the next scene representation from discrete visual tokens autoregressively. During training, HybridDriveVLA minimizes the divergence between the predicted $\hat{I}_{t+6\alpha}$ and a ground-truth $I_{t+6\alpha}$. Each predicted next scene image $\hat{I}_{t+6\alpha}$ is represented as a sequence of d discrete tokens, where $\sigma = 512$ is the fixed dimension of the VQGAN Encoder. Let $\{q_1, \ldots, q_\sigma\}$ be the sequence of VQGAN-encoded visual tokens representing a ground-truth future scene $\hat{I}_{t+6\alpha}$. Given the preceding tokens the model predicts the conditional probability of the next token using M. The training objective for the visual reasoning V-CoT is to minimise $\mathcal{L}_{V\text{-CoT}}$ defined as:

$$\mathcal{L}_{V\text{-CoT}} = -\sum_{d=1}^{\sigma} \log P_\theta(q_d \mid q_{<\sigma}, I_t, E, l_t, c_t, o_t), \tag{2}$$

where q_d denotes the ground-truth visual token at position d, and θ represents HybridDriveVLA parameters. Through this process, HybridDriveVLA learn visual next scenes images prediction and maintain temporal coherence across visual scenes.

3.3. Tree-of-Thought-Evaluation (ToT-Evaluation)

While V-CoT anticipates a visual goal image $\hat{I}_{t+6\alpha}$, ToT-Evaluation deliberates on the optimal sequence of waypoints to achieve it. Instead of predicting a single path, this module generates a set of N candidate sequences of waypoints. Let $\mathcal{A}^n = \{a^n_{t+\alpha}, a^n_{t+2\alpha}, \ldots, a^n_{t+6\alpha}\}$ be the n-th sequence of waypoints, where $n \in \{1, \ldots, N\}$. For each waypoint $a^n_{t+k\alpha}$ within a sequence, HybridDriveVLA also predicts corresponding aspects set of scores $\mathcal{S}^n_{t+k\alpha}$. The generation of all sequences and the scores of each waypoint is conditioned on the current state and a visual scene goal provided by V-CoT:

$$\{(a^n_{t+k\alpha}, \mathcal{S}^n_{t+k\alpha})\}_{n=1}^{N} = M(I_t, \hat{I}_{t+6\alpha}, E, l_t, c_t, o_t) \quad (3)$$

where $\mathcal{S}^n$ represents the set of scores for a unique waypoint $a^n_{t+k\alpha}$ in the sequence $\mathcal{A}^n$. During training, HybridDriveVLA learns to predict these sequences and scores. At inference time, the optimal sequence of waypoints, $\mathcal{A}^*$, is selected by identifying waypoints with the highest cumulative score across all aspects and time steps. We define a total score of a waypoint at k as:

$$T^n_{t+k\alpha} = \sum_j s^{n,j}_{t+k\alpha} \quad (4)$$

with $j \in \{\text{safety}, \text{progress}, \text{comfort}\}$. The index of the optimal waypoint, n^* at $t + k\alpha$ is then obtained by:

$$n^* =_{n\in\{1,\ldots,N\}} T^n_{t+k\alpha} \quad (5)$$

This deliberation process, which is a reasoning-based beam search [15] where each node is a waypoint, enables HybridDriveVLA to reason beyond single-path to adaptively evaluate and select the most efficient waypoints. The training objective for this module, $\mathcal{L}_{\text{ToT-Eval}}$, maximizes the log-probability of generating the ground-truth sequences of waypoints and each of these waypoints scores autoregressively. The objective is formulated as:

$$\mathcal{L}_{\text{ToT-Eval}} = -\sum_{n=1}^{N}\sum_{k=1}^{6}\sum_{j} \log P_\theta\big(a^n_{t+k\alpha}, s^{n,j}_{t+k\alpha} \,\big|\, a^n_{<t+k\alpha}, s^{n,j}_{<t+k\alpha}, I_t, \hat{I}_{t+6\alpha}, E, l_t, c_t, o_t\big) \quad (6)$$

This ensures that the predicted waypoints and their evaluations are temporally consistent. The total training objective for HybridDriveVLA combines both reasoning losses:

$$\mathcal{L}_{HybridDriveVLA} = \mathcal{L}_{\text{V-CoT}} + \mathcal{L}_{\text{ToT-Eval}} \quad (7)$$

This composite objective jointly optimizes visual prediction and evaluative planning, ensuring that the anticipated goals and the sequences of waypoints remain consistent.

3.4. Supervised Fine-Tuning

HybridDriveVLA is first trained through Supervised Fine-Tuning (SFT) to solidify its foundational visual reasoning and predictive abilities. Initialized from a large-scale VLA model, HybridDriveVLA is fine-tuned on paired samples of multi-view inputs, scene annotations, and corresponding next scene images. During SFT, both the next scene prediction objective and multiple waypoint sequences are optimized jointly. This dual-stage optimization enables HybridDriveVLA to learn causal visual-to-action dependencies and strengthen temporal coherence between scene synthesis and trajectory planning. During this phase, HybridDriveVLA is optimized to enhance the next scene generation and planning by minimizing $\mathcal{L}_{HybridDriveVLA}$ This unified design ensures consistent grounding between perception, prediction, and planning within a single generative reasoning stream. This process enables accurate reconstruction of lane layouts, object positions, and temporally coherent next-frame scenes. It also aligns the visual and linguistic token spaces, allowing HybridDriveVLA to reason spatially using the same sequential logic applied in text-based generative models.

3.5. Instruction Tuning

After SFT, HybridDriveVLA undergoes Instruction Tuning to enhance interpretability, reasoning flexibility, and deliberation. HybridDriveVLA is trained on a visual-instruction corpus derived from nuScenes driving datasets, where each instance pairs a visual context with natural-language prompts guiding multi-step reasoning and evaluation. By conditioning our model on such prompts, HybridDriveVLA learns to generate reasoning-consistent multiple waypoint sequences and generate next scenes by minimizing $\mathcal{L}_{HybridDriveVLA}$. This phase refines the ToT-Evaluation's performance, teaching HybridDriveVLA to refine its internal deliberations and to maintain coherent reasoning across time. While SFT teaches HybridDriveVLA to predict and plan through visual anticipation, Instruction Tuning extends this capability to reasoning under each aspect, transforming visual-driven prediction into deliberative planning

3.6. Dataset Construction

To effectively train and evaluate the hierarchical reasoning capabilities of HybridDriveVLA, we construct two complementary datasets derived from the large-scale nuScenes [1] benchmark. Each dataset is designed to support a specific training stage, corresponding respectively to supervised-fine-tuning for visual generation and instruction-tuning reasoning.

nuScenes Overview. The nuScenes dataset provides rich multimodal driving data, including 1,000 driving sequences (each approximately 20 seconds long) captured using a 360°

camera suite and LiDAR sensors. It contains 28,130 training and 6,019 validation samples, each annotated with ego-vehicle states, object bounding boxes, and scene-level metadata. We leverage this dataset to construct training pairs that align with the reasoning and evaluative objectives of HybridDriveVLA.

Supervised-Fine-Tuning Visual Generation Dataset. This dataset supports the Visual Chain-of-Thought (V-CoT) training phase, which focuses on autoregressive future scene generation. For each temporal pair $(t, t+6\alpha)$, we extract the multi-view camera frames I_t and their corresponding future frame $I_{t+6\alpha}$, where $t+6\alpha$ denotes the prediction horizon. Each future scene image is tokenized via a pre-trained MoVQGAN encoder [30], producing discrete visual tokens. Training samples take the form of prompt–response pairs where the input prompt includes $\hat{I}_t$ and a concise instruction (e.g.,*"Given ego position, speed, previous trajectory, actual scene image, Generate the next scene view 3s seconds ahead."*), while the response is a serialized sequence of ground truth visual tokens for $I_{t+6\alpha}$. This stage allows HybridDriveVLA to internalize visual causality and learn temporally consistent scene evolution.

Instruction-Tuning Reasoning Dataset. We construct an instruction tuning dataset by templating visual-language pairs with scene descriptions and aspect evaluation prompts derived from nuScenes metadata. This dataset targets the joint reasoning and deliberation abilities central to HybridDriveVLA's architecture. Each sample contains a structured prompt combining the current camera view I_t and recent ego-vehicle trajectory history, a navigational command (LEFT, RIGHT, FORWARD), and aspects E specifying driving priorities and the predicted future scene $\hat{I}_{t+6\alpha}$ The corresponding response contain ground truth N sequences of waypoint at each $t+k\alpha$ associate with scores for safety, progress, and comfort. Each waypoint in the response is annotated with its ground-truth aspect score vector $\mathbf{s}_{t+k\alpha}$ computed in Section 3 following prior work on nuScenes [8], ensuring interpretable supervision. To generate the ground truth scores for training the ToT-Evaluation module, we normalize the ground truth waypoints at each step $t+k\alpha$ based on the nuScene dataset statistics. Let $s^{safety}_{t+k\alpha}$, $s^{comfort}_{t+k\alpha}$, and $s^{progress}_{t+k\alpha}$ be the ground truth scores at $t+k\alpha$. These are computed as follows:

- **Safety score:**At each time $t+k\alpha$, let $d_{t+k\alpha}$ be the minimum Euclidean distance from the ego-vehicle to any other object. Over the dataset, d^{min} and d^{avg}. The score $s^{safety}_{t+k\alpha}$ is normalized linearly, mapping d^{min} to the lowest safety score and d^{avg} to the maximum safety:

$$s^{\text{safety}}_{t+k\alpha} = \sigma\left(\frac{d_{t+k\alpha} - d^{\text{min}}}{d^{\text{avg}} - d^{\text{min}}}\right) \tag{8}$$

  Where higher values corresponding to high safety.
- **Comfort score:**At each time $t+k\alpha$, let $c_{t+k\alpha}$ denote the steering rate of the ego-vehicle. Across the dataset, define c^{min} and c^{avg}. The comfort score $s^{\text{comfort}}_{t+k\alpha}$ is normalized such that c^{min} maps to the highest comfort and c^{avg} to the lowest comfort:

$$s^{\text{comfort}}_{t+k\alpha} = \sigma\left(1 - \frac{c_{t+k\alpha} - c^{\text{min}}}{c^{\text{avg}} - c^{\text{min}}}\right) \tag{9}$$

  Where higher values corresponding to smoother and more comfortable driving dynamics.
- **Progress score:** At each time $t+k\alpha$, let $v_{t+k\alpha}$ be the speed of the ego-vehicle. Over the dataset, let v^{min} and v^{avg}. The progress score $s^{\text{progress}}_{t+k\alpha}$ is normalized such that v^{min} maps to the lowest progress and v^{avg} to the highest progress:

$$s^{\text{progress}}_{t+k\alpha} = \sigma\left(\frac{v_{t+k\alpha} - v^{\text{min}}}{v^{\text{avg}} - v^{\text{min}}}\right) \tag{10}$$

These scores are used at each instant as the ground-truth targets for the score $s^{n,j}_{t+k\alpha}$ that the ToT-Evaluation learns to predict. This dataset enables HybridDriveVLA to align predictive visual reasoning with multi-aspect evaluation.

4. Experiments

Evaluation Metrics: We report the following metrics under both the per-timestep style of UniAD [8] and the averaged style of ST-P3 [7] for comprehensive comparison.

- L2 Displacement Error (m): The Euclidean distance between the predicted and ground-truth waypoints.
- Collision Rate (%): The percentage of predicted trajectories that result in a collision with scene obstacles.
- UniAD Metrics: We evaluate planning performance with per-timestep and overall L2 error and collision rates, using the official validation protocol.
- ST-P3 Metrics: We compute the average L2 error and collision rate across all future waypoints for a fair comparison with prior work.

For the NAVSIM benchmark, we report the standard metrics, where higher is better for all:

- PDMS: The Planning Diversity and Motion Sickness score, which evaluates the quality and realism of the driving behavior.
- Collision: The collision-free rate.
- Area, Direction, Progress: Metrics measuring adherence to drivable areas, heading, and goal progress.
- TTC: Time-to-Collision, a key safety metric.
- Comfort: A score reflecting the smoothness of the trajectory (e.g., low jerk and acceleration).

Implementation Details: Our HybridDriveVLA was built upon the Qwen2-VL-2B model [22]. During supervised fine-tuning our model was trained for 20 epochs with a learning rate of 1.0×10^{-4}, a batch size of 8 per device, and

Method	ST-P3 Metrics								UniAD Metrics								Model
	L2 (m) ↓				Collision (%) ↓				L2 (m) ↓				Collision (%) ↓				
	1s	2s	3s	Avg.	1s	2s	3s	Avg.	1s	2s	3s	Avg.	1s	2s	3s	Avg.	
Non-Autoregressive Methods																	
ST-P3 [7][ECCV 2022]	1.33	2.11	2.90	2.11	0.23	0.62	1.27	0.71	–	–	–	–	–	–	–	–	–
VAD [9][ICCV 2023]	0.17	0.34	0.60	0.37	0.07	0.10	0.24	0.14	–	–	–	–	–	–	–	–	–
UniAD [8][CVPR 2023]	0.44	0.67	0.96	0.69	0.04	0.08	0.23	0.12	0.48	0.96	1.65	1.03	0.05	0.17	0.71	0.31	–
GenAD [31][CVPR 2024]	0.28	0.49	0.78	0.52	0.08	0.14	0.34	0.19	–	–	–	–	–	–	–	–	
Autoregressive Methods																	
OminiDrive [24][CVPR 2025]	0.40	0.80	1.32	0.84	0.04	0.46	2.32	0.94	–	–	–	–	–	–	–	–	LLaVA-7B
DriveVLM [23][arXiv 2024]	0.18	0.34	0.68	0.40	0.10	0.22	0.45	0.27	–	–	–	–	–	–	–	–	Qwen-VL-7B Visual
RDA-Driver [18][ECCV 2024]	0.17	0.37	0.69	0.40	**0.01**	**0.05**	0.26	**0.10**	0.23	0.73	1.54	0.80	0.00	0.13	0.83	0.32	LLaVa-7B
GPT-Driver [13][NeurIPS 2023]	0.20	0.40	0.70	0.44	0.04	0.12	0.36	0.17	0.27	0.74	1.52	0.84	0.07	0.15	1.10	0.44	GPT-3.5
OpenDriveVLA [32][arxiv 2024]	**0.14**	0.30	0.55	0.33	0.02	0.07	**0.22**	**0.10**	0.19	0.58	1.24	0.67	**0.02**	0.18	0.70	0.30	Qwen2.5-3B
HybridDriveVLA (ToT-Evaluation)	**0.14**	0.37	0.41	0.30	0.09	0.12	0.27	0.16	0.23	0.38	0.61	0.40	0.09	0.25	0.57	0.30	Qwen2-VL-2B
HybridDriveVLA(Full)	0.19	**0.24**	**0.36**	**0.26**	0.04	0.16	0.24	0.14	**0.18**	**0.34**	**0.43**	**0.31**	0.06	**0.08**	**0.45**	**0.19**	Qwen2-VL-2B

Table 1. Trajectory planning on the nuScenes validation set.

Method	PDMS ↑	Collision ↑	Area ↑	Direction ↑	Progress ↑	TTC ↑	Comfort ↑
TransFuser [3]	83.88	97.78	92.63	97.97	78.88	92.89	99.98
DRAMA [12]	86.87	98.19	95.18	**98.03**	81.33	94.17	**100.00**
Hydra-MDP [11]	91.26	99.07	98.29	95.79	85.20	96.56	**100.00**
Centaur [21]	92.10	99.23	**98.72**	96.77	85.96	97.17	99.97
TrajHF [10]	93.95	99.30	97.51	91.72	90.39	98.02	99.81
AutoVLA [33][NeurIPS 2025]	92.12	99.14	97.08	95.51	87.55	97.12	99.98
HybridDriveVLA (Safety)	**94.89**	**99.62**	97.69	97.50	**92.74**	98.73	97.91
HybridDriveVLA (Comfort)	91.73	92.75	96.41	97.31	91.34	94.18	97.72
HybridDriveVLA (Progress)	93.12	90.75	93.41	97.31	96.82	97.18	92.53
HybridDriveVLA (Optimal)	94.62	97.63	96.38	95.43	96.56	**99.42**	96.85

Table 2. Testing Results on the NAVSIM Benchmark

16 gradient accumulation steps to learn visual token generation. Then during instruction tuning, we trained HybridDriveVLA for 20 epochs with the same configuration using LoRA [6]. Key hyperparameters included a waypoint prediction interval α of 0.5s, and a cosine learning rate scheduler. The vision tower of the backbone Language model was kept frozen throughout this instruction tuning to preserve its learned features, while the language model and LoRA adapters were updated. For visual tokenization, we utilized a pre-trained MoVQGAN.

4.1. Main Results

Table 1 showed the trajectory planning performance of HybridDriveVLA on the nuScenes validation dataset. Our model demonstrated state-of-the-art performance among language model-based autoregressive methods, particularly in reducing collision rates. When compared to other autoregressive models like DriveVLM and OpenDriveVLA, HybridDriveVLA achieved a significant reduction in collisions under both ST-P3 and UniAD metrics. For instance, our final model, HybridDriveVLA (CoT + ToT), achieved an average collision rate of 0.17% (ST-P3) and 0.19% (UniAD). This result is highly competitive even with non-autoregressive methods and highlights the effectiveness of our V-CoT and ToT-Evaluation reasoning, which allowed the HybridDriveVLA to anticipate next scene images and deliberate on optimal trajectories. The low L2 error further confirmed the planning accuracy.

4.2. Ablation Studies

We conducted ablation studies to validate the core components of HybridDriveVLA. All ablations used UniAD-style metrics and excluded ego vehicle states unless specified. Table 2 showcased our model's ability to adapt its driving style based on the prioritized aspect of the NAVSIM benchmark. When guided to prioritize Safety, the model achieved the highest Time-to-Collision (TTC) score of 98.73, indicating safer behavior. When focused on comfort, it yields the best Comfort score of 97.72. The Optimal sequence, which balances all aspects, achieved a strong overall PDMS score of 94.62 and a high collision-free rate of 97.63. This confirms that our ToT-Evaluation successfully evaluates the waypoints in an interpretable manner. To quantify the impact of our reasoning components, we analyzed the results in Table 1. HybridDriveVLA (ToT-Evaluation) model, which performs deliberation without visual anticipation from V-CoT, achieves an average collision rate of 0.23% on the ST-P3 benchmark. By integrating V-CoT, our



Figure 4. Qualitative analysis of HybridDriveVLA on the nuScenes dataset. The red trajectory is the predicted optimal sequence of waypoints, and the green is the ground truth.

full HybridDriveVLA model reduced this collision rate to 0.17%, a relative improvement of 26%. This result empirically proved that while deliberative evaluation was effective on its own, its performance has significantly enhanced when grounded to an anticipated future visual scene. It confirmed that V-CoT and ToT-Evaluation are complementary and crucial for achieving efficiency.

Qualitative Analysis. In Table 1, HybridDriveVLA (v-ToT), which represents training without the initial SFT stage for V-CoT, has a significantly higher collision rate of 0.23% for ST-P3 compared to the full HybridDriveVLA that had 0.17% for ST-P3. This demonstrated that equipping the model with an understanding of visual scenes via SFT was critical before it can effectively reason and deliberate during instruction tuning. Figure 4 provided qualitative examples of HybridDriveVLA's deliberative process. In Figure 4 scenario, a direct path following the command without deliberative evaluation leads to a potential conflict with another vehicle. The baseline model, lacking deliberation, committed to this dangerous trajectory. In contrast, HybridDriveVLA's ToT-Evaluation generated and evaluated multiple trajectories and correctly assigned a lower score to the unsafe, uncomfortable and unprogressive waypoints and selected an optimal sequence of waypoints, showcasing its ability to navigate complex interactions through deliberative reasoning.

5. Conclusion

In this paper, we introduced HybridDriveVLA, a novel Vision-Language-Action model designed to address limitations in current autonomous driving models. We identified that traditional VLA models relied on Text-to-Image reasoning, leading to visual information under-usage. They also just predict a single sequence of waypoints as a trajectory, considering simultaneously multiple aspects. To overcome these challenges, we proposed a paradigm shift inspired by human cognition, integrating in a baseline VLA model a Visual Chain-of-Thought (V-CoT) and a ToT-inspired evaluation (ToT-Evaluation) to reveal the importance of each aspect. Through extensive experiments on the nuScenes benchmark, we demonstrated that HybridDriveVLA achieves state-of-the-art performance among autoregressive VLA models. Most notably, it reduces the average collision rate to 0.17%, confirming that the generation of visual next scenes as goal and a deliberative evaluation to reveal the importance of driving aspects is crucial for autonomous driving systems. Our ablation studies further validated the effectiveness of our approach, confirming that the integration of V-CoT and ToT-Evaluation improves the reliability of autonomous driving systems.

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